Linear Systems
1. Introduction to Linear Systems

- **Linear Equation**: An equation of the form $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$.
  - **Homogeneous**: $b = 0$ (e.g., $2x + 3y = 0$).
  - **Standard Form**: Variables on left, constants on right.
- **Linear System**: A collection of linear equations.
  Example:

  $$\begin{cases} x + y = 2 \\ x - y = 0 \end{cases}$$

  Solution: $x = 1, y = 1$ (intersection of two lines).

Types of Solutions

1. **No Solution** (Inconsistent): Parallel lines/planes.
2. **Unique Solution**: Lines/planes intersect at one point.
3. **Infinitely Many Solutions**: Lines/planes coincide.

jkhsdjbksdfkbjsdfkjbajnansn

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2. Matrices and Augmented Matrices

- **Augmented Matrix**: Represents a linear system by combining coefficients and constants.
  Example:
  Given the linear system:

  $$\begin{cases} 3x_1 + 2x_2 - x_3 = 1 \\ 5x_2 + x_3 = 3 \\ x_1 + x_3 = 2 \end{cases}$$

  The corresponding augmented matrix is:

  $$\left[\begin{array}{ccc|c} 3 & 2 & -1 & 1 \\ 0 & 5 & 1 & 3 \\ 1 & 0 & 1 & 2 \end{array}\right]$$

- **Structure**:
  - Left of the vertical bar: Coefficients of variables (x_1, x_2, x_3).
  - Right of the vertical bar: Constants $(1, 3, 2)$.
- **Homogeneous System**: Augmented matrix with the last column (constants) all zeros.
  Example:

  $$\left[\begin{array}{ccc|c} 2 & -1 & 4 & 0 \\ 0 & 3 & 1 & 0 \end{array}\right]$$

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3. Row-Echelon Form (REF) and Reduced REF (RREF)

Row-Echelon Form (REF)

1. **Zero rows** at the bottom.
2. **Leading entry** (first non-zero entry in a row) is to the right of the row above.

Example 1 (REF):

$$\begin{pmatrix} \color{red}{1} & 2 & 3 \\ 0 & \color{red}{3} & 1 \\ 0 & 0 & \color{red}{2} \end{pmatrix}$$

- Red entries: Leading entries (staircase pattern).
- Non-zero rows above zero rows.

Example 2 (REF):

$$\begin{pmatrix} \color{red}{2} & 5 & -1 & 4 \\ 0 & 0 & \color{red}{3} & 1 \\ 0 & 0 & 0 & \color{red}{5} \end{pmatrix}$$

- Leading entries (red) shift right in subsequent rows.

Reduced Row-Echelon Form (RREF)

1. **Leading entries are 1** (highlighted in red).
2. **Zeros above/below** leading 1s in pivot columns.

Example 1 (RREF):

$$\begin{pmatrix} \color{red}{1} & 0 & 2 \\ 0 & \color{red}{1} & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

- Red 1s: Leading entries.
- Zeros in all positions of pivot columns except leading 1s.

Example 2 (RREF):

$$\begin{pmatrix} \color{red}{1} & 0 & 0 & 4 \\ 0 & \color{red}{1} & 0 & -2 \\ 0 & 0 & \color{red}{1} & 3 \end{pmatrix}$$

- Each leading 1 (red) is the only non-zero entry in its column.

Pivot Columns vs. Free Variables

- **Pivot Columns**: Columns containing leading entries (red).
- **Non-Pivot Columns**: Columns without leading entries; correspond to free variables.

Example:
For the RREF matrix:

$$\begin{pmatrix} \color{red}{1} & 2 & 0 & 3 \\ 0 & 0 & \color{red}{1} & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

- Pivot columns: Column 1 (red 1) and Column 3 (red 1).
- Free variables: x_2 and x_4 (non-pivot columns).

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4. Elementary Row Operations

Three types:

1. **Row Swap**: $R_i \leftrightarrow R_j$.
2. **Row Scaling**: $kR_i \to R_i$ $(k \neq 0)$.
3. **Row Replacement**: $R_i + kR_j \to R_i$.

Key Property: Row operations do not change the solution set.

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5. Gaussian Elimination

Steps to solve a linear system:

1. **Forward Elimination**: Convert to REF.
2. **Back-Substitution**: Solve from bottom up.

Gaussian Elimination Example

System:

$$\begin{cases} x+y-z=-2 \\ 2x-y+z=5 \\ -x+2y+2z=1 \end{cases}$$

Step 1: Write the Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 2 & -1 & 1 & 5 \\ -1 & 2 & 2 & 1 \end{array}\right]$$

Step 2: Forward Elimination to REF

Step 2.1: Eliminate x from Row 2 and Row 3

- **Row 2**: $R2 \leftarrow R2 - 2R1$

$$\text{New Row 2} = (0, -3, 3, 9)$$

- **Row 3**: $R3 \leftarrow R3 + R1$

$$\text{New Row 3} = (0, 3, 1, -1)$$

Updated Matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & -3 & 3 & 9 \\ 0 & 3 & 1 & -1 \end{array}\right]$$

Step 2.2: Eliminate y from Row 3

- **Row 3**: $R3 \leftarrow R3 + R2$

$$\text{New Row 3} = (0, 0, 4, 8)$$

Final REF Matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & -3 & 3 & 9 \\ 0 & 0 & 4 & 8 \end{array}\right]$$

Step 3: Back-Substitution

Step 3.1: Solve for z

From Row 3:

$$4z = 8 \implies z = 2$$

Step 3.2: Solve for y

From Row 2:

$$-3y + 3z = 9 \implies -3y + 6 = 9 \implies y = -1$$

Step 3.3: Solve for x

From Row 1:

$$x + y - z = -2 \implies x - 1 - 2 = -2 \implies x = 1$$

Final Solution

$$(x, y, z) = (1, -1, 2)$$

Verification

Substitute into original equations:

1. $1 + (-1) - 2 = -2$ ✓
2. $2(1) - (-1) + 2 = 5$ ✓
3. $-1 + 2(-1) + 2(2) = 1$ ✓

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6. Gauss-Jordan Elimination

- Extends Gaussian Elimination to RREF.
- **Steps**:
  1. Convert to REF.
  2. Scale rows to make pivots 1.
  3. Eliminate entries above pivots.

Gauss-Jordan Elimination Example

System:

$$\begin{cases} x + y - z = 7 \\ x - y + 2z = 3 \\ 2x + y + z = 9 \end{cases}$$

Step 1: Write the Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 7 \\ 1 & -1 & 2 & 3 \\ 2 & 1 & 1 & 9 \end{array}\right]$$

Step 2: Forward Elimination to REF

Step 2.1: Eliminate x from Row 2 and Row 3

- **Row 2**: $R2 \leftarrow R2 - R1$

$$\text{New Row 2} = (0, -2, 3, -4)$$

- **Row 3**: $R3 \leftarrow R3 - 2R1$

$$\text{New Row 3} = (0, -1, 3, -5)$$

Updated Matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 7 \\ 0 & -2 & 3 & -4 \\ 0 & -1 & 3 & -5 \end{array}\right]$$

Step 2.2: Normalize Row 2

- **Row 2**: $R2 \leftarrow -\frac{1}{2}R2$

$$\text{New Row 2} = (0, 1, -\frac{3}{2}, 2)$$

Updated Matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 7 \\ 0 & 1 & -\frac{3}{2} & 2 \\ 0 & -1 & 3 & -5 \end{array}\right]$$

Step 2.3: Eliminate y from Row 3

- **Row 3**: $R3 \leftarrow R3 + R2$

$$\text{New Row 3} = (0, 0, \frac{3}{2}, -3)$$

Updated Matrix (REF):

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 7 \\ 0 & 1 & -\frac{3}{2} & 2 \\ 0 & 0 & \frac{3}{2} & -3 \end{array}\right]$$

Step 3: Backward Elimination to RREF

Step 3.1: Normalize Row 3

- **Row 3**: $R3 \leftarrow \frac{2}{3}R3$

$$\text{New Row 3} = (0, 0, 1, -2)$$

Updated Matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 7 \\ 0 & 1 & -\frac{3}{2} & 2 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

Step 3.2: Eliminate z from Rows 1 and 2

- **Row 2**: $R2 \leftarrow R2 + \frac{3}{2}R3$

$$\text{New Row 2} = (0, 1, 0, -1)$$

- **Row 1**: $R1 \leftarrow R1 + R3$

$$\text{New Row 1} = (1, 1, 0, 5)$$

Updated Matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

Step 3.3: Eliminate y from Row 1

- **Row 1**: $R1 \leftarrow R1 - R2$

$$\text{New Row 1} = (1, 0, 0, 6)$$

Final RREF:

$$\left[\begin{array}{ccc|c} \color{red}{1} & 0 & 0 & 6 \\ 0 & \color{red}{1} & 0 & -1 \\ 0 & 0 & \color{red}{1} & -2 \end{array}\right]$$

Step 4: Interpret the Solution

From the RREF matrix:

- $x = 6$

- $y = -1$
- $z = -2$

Unique Solution:

$$(x, y, z) = (6, -1, -2)$$

Verification

Substitute into original equations:

1. $6 + (-1) - (-2) = 7$ √
2. $6 - (-1) + 2(-2) = 3$ √
3. $2(6) + (-1) + (-2) = 9$ √

Properties of a Matrix in Reduced Row-Echelon Form (RREF)

Let R be an $n \times m$ matrix in **reduced row-echelon form (RREF)**. The following properties hold:

1. Pivot Columns and Nonzero Rows

- Each nonzero row contains exactly **one leading 1** (pivot), and these pivots appear in different columns.
- **The number of pivot columns is equal to the number of nonzero rows.**

2. Non-pivot Columns and Zero Rows

- The **number of non-pivot columns** is given by $m - r$, where r is the number of pivot columns (or nonzero rows).
- The **number of zero rows** is given by $n - r$.
- In general, the number of non-pivot columns **is not necessarily equal** to the number of zero rows.

3. Common Misconceptions

- The **number of non-pivot columns is NOT necessarily equal** to the number of nonzero rows.
- The **number of pivot columns is NOT necessarily equal** to the number of zero rows.

Summary

Statement	True/False
The number of pivot columns = the number of nonzero rows	✅ True
The number of nonpivot columns = the number of zero rows	X False
The number of nonpivot columns = the number of nonzero rows	X False
The number of pivot columns = the number of zero rows	X False

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7. Solutions of Linear Systems

General Solution

- **Free Variables**: Assigned parameters (e.g., s, t).
- **Example**:

$$\begin{cases} x + 2y = 5 \\ 2x + 4y = 10 \end{cases}$$

  General solution: $x = 5 - 2s$, $y = s$ $(s \in \mathbb{R})$.

Homogeneous Systems

- Always has the trivial solution $x_1 = x_2 = \cdots = 0$.
- Non-trivial solutions exist if there are free variables.

Solving a Linear System with Unknown Coefficient a

System:

$$\begin{cases} ax + y - z = 1 \\ x + y + 2z = 3 \\ x + (a-1)y - z = 0 \end{cases}$$

Step 1: Augmented Matrix

$$\left[\begin{array}{ccc|c} a & 1 & -1 & 1 \\ 1 & 1 & 2 & 3 \\ 1 & a-1 & -1 & 0 \end{array}\right]$$

Step 2: Case Analysis for a

Case 1: $a = 1$

Simplified System:

$$\begin{cases} x + y - z = 1 \\ x + y + 2z = 3 \\ x - z = 0 \end{cases}$$

Augmented Matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 1 & 1 & 2 & 3 \\ 1 & 0 & -1 & 0 \end{array}\right]$$

Row Operations:

1. $R2 \leftarrow R2 - R1$:

$$\text{New Row 2} = (0, 0, 3, 2)$$

2. $R3 \leftarrow R3 - R1$:

$$\text{New Row 3} = (0, -1, 0, -1)$$

3. Swap $R2$ and $R3$:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 3 & 2 \end{array}\right]$$

   Back-Substitution:

- $z = \frac{2}{3}$
- $y = 1$
- $x = \frac{2}{3}$

Solution:

$$(x, y, z) = \left(\frac{2}{3}, 1, \frac{2}{3}\right)$$

Case 2: $a = 0$

Simplified System:

$$\begin{cases} y - z = 1 \\ x + y + 2z = 3 \\ x - y - z = 0 \end{cases}$$

Augmented Matrix:

$$\left[\begin{array}{ccc|c} 0 & 1 & -1 & 1 \\ 1 & 1 & 2 & 3 \\ 1 & -1 & -1 & 0 \end{array}\right]$$

Row Operations:

1. Swap $R1$ and $R2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & -1 & 1 \\ 1 & -1 & -1 & 0 \end{array}\right]$$

2. $R3 \leftarrow R3 - R1$:

$$\text{New Row 3} = (0, -2, -3, -3)$$

3. $R3 \leftarrow R3 + 2R2$:

$$\text{New Row 3} = (0, 0, -5, -1)$$

Back-Substitution:

- $z = \frac{1}{5}$
- $y = \frac{6}{5}$
- $x = \frac{7}{5}$

Solution:

$$(x, y, z) = \left(\frac{7}{5}, \frac{6}{5}, \frac{1}{5}\right)$$

General Case: $a \neq 0, 1$

Proceed with Gaussian elimination carefully:

1. Eliminate x from $R2$ and $R3$.
2. Simplify rows while avoiding division by a.
3. Express variables in terms of a.

Key Insight:
The system has a unique solution for $a \neq 0, 1$, no solution for certain a, or infinitely many solutions depending on a.

Final

- **Unique Solution**: For $a \neq 0, 1$, solve using standard elimination.
- **Inconsistent System**: Check for contradictions after substitution.
- **Free Variables**: If elimination leads to dependent rows, assign parameters to free variables.

Verification: Substitute solutions back into original equations to confirm validity.

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8. Applications

Balancing Chemical Equations

Example: Balance $CH_4 + O_2 \rightarrow CO_2 + H_2O$.

- System:

$$
\begin{cases} x_1 - x_3 = 0 \\ 4x_1 - 2x_4 = 0 \\ 2x_2 - 2x_3 - x_4 = 0 \end{cases}
$$

- Solution: $x_1 = 1$, $x_2 = 2$, $x_3 = 1$, $x_4 = 2$.

Traffic Flow Analysis

- Junction equations (inflow = outflow).
- Example: If $x_6 = 50$ and $x_7 = 100$, solve for $x_1, \ldots, x_5$.

Temperature Distribution

- Interior node temperature = average of neighbors.
- Example: $T_1 = \frac{60+100+T_2+T_3}{4} \Rightarrow 4T_1 - T_2 - T_3 = 160$.

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9. Key Theorems

1. A linear system has **no solution**, **one solution**, or **infinitely many solutions**.
2. A system is **inconsistent** iff the RHS column is a pivot column in REF.
3. The number of free variables = number of non-pivot columns.

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10. MATLAB Applications

1. Solving Linear Systems with `rref()`

Example 1: Inconsistent System

System:

$$
\begin{cases} 3x_1 + 5x_2 - x_3 = 11 \\ x_1 + x_2 - x_3 = -1 \\ 5x_1 + 7x_2 - 3x_3 = 13 \end{cases}
$$

MATLAB Code:

```
% Define the augmented matrix
A = [3, 5, -1, 11;
     1, 1, -1, -1;
     5, 7, -3, 13];

% Compute RREF
rref_A = rref(A);

disp('Reduced Row Echelon Form:');
disp(rref_A);
```

Output Interpretation:

$$
\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]
$$

- **Inconsistent System**: The last row implies $0x_1 + 0x_2 + 0x_3 = 1$, which is impossible.

Example 2: Infinitely Many Solutions

System:

$$\begin{cases} 3x - y + 2z = 0 \\ 3x + 3y - z = 1 \\ 9x + y + 3z = 1 \end{cases}$$

MATLAB Code:

```
% Define the augmented matrix
A = [3, -1, 2, 0;
     3, 3, -1, 1;
     9, 1, 3, 1];

% Compute RREF
rref_A = rref(A);

disp('Reduced Row Echelon Form:');
disp(rref_A);
```

Output:

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{5}{12} & \frac{1}{12} \\ 0 & 1 & -\frac{3}{4} & \frac{1}{4} \\ 0 & 0 & 0 & 0 \end{array}\right]$$

- **General Solution**: Let $z = t$:
  - $x = \frac{1}{12} - \frac{5}{12}t$
  - $y = \frac{1}{4} + \frac{3}{4}t$
  - $z = t$

2. Visualizing Temperature Distributions

Problem: Estimate temperatures T_1, T_2, T_3, T_4 at interior nodes of a square plate using the averaging rule.
Given:

- Edge temperatures: 60°C (left), 100°C (right), 80°C (top), 40°C (bottom).
- Interior node equations (e.g., $T_1 = \frac{60+100+T_2+T_3}{4}$).

Step 1: Construct the Linear System

$$\begin{cases} 4T_1 - T_2 - T_3 = 160 \\ -T_1 + 4T_2 - T_4 = 80 \\ -T_1 + 4T_3 - T_4 = 80 \\ -T_2 - T_3 + 4T_4 = 40 \end{cases}$$

MATLAB Code:

```
% Coefficient matrix and constants
A = [4, -1, -1, 0;
     -1, 4, 0, -1;
     -1, 0, 4, -1;
      0, -1, -1, 4];
b = [160; 80; 80; 40];

% Solve using matrix inversion
T = A \ b;

% Display temperatures
disp('Temperatures:');
disp(T);
```

Step 2: Visualize with Mesh Grid

```
% Create grid
[X, Y] = meshgrid(1:3, 1:3);
```

```matlab
% Map temperatures to grid points
% Assume T1, T2, T3, T4 correspond to positions (2,2), (2,1), (2,3), (1,2)
temp_grid = [80, 100, 80;
             60, T(1), 40;
             80, 60, 80];

% Plot
surf(X, Y, temp_grid);
xlabel('X-axis');
ylabel('Y-axis');
zlabel('Temperature (°C)');
title('Temperature Distribution on Square Plate');
```

Output:

- A 3D surface plot showing temperature gradients across the plate.

Key MATLAB Commands

1. `rref(A)`: Computes the reduced row echelon form of matrix `A`.
2. `A \ b`: Solves the linear system $Ax = b$ using matrix inversion.
3. `meshgrid`/`surf`: Creates grids and plots 3D surfaces.